The Long Boil: Game Design Document

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A steampunk idle game

1. Executive Summary

The Long Boil is a steampunk idle MMO for web and phone: 10 non-combat and 7 combat skills, a player-run market, guilds, and AFK party combat. You are a Tinker in Kettleport, a floating city, and work continues for 12 hours offline, so a session is a 3 to 5 minute check-in.

- **Win condition:** none in the usual sense, since an idle game never ends; success means higher levels, better and higher-calibrated gear, and leaderboard ranks (total level, each skill, net worth, guilds).
- **Loss states:** a knockout (60-second respawn, then the queue resumes), a lost boss fight (the attempt is spent), or a failed calibration (gear falls to +0 unless protected).
- **Business:** free to play; Stripe-sold Gilded Cogs buy time savers and cosmetics, and nothing is exclusive to paying players.

2. Game Mechanics

Core loop. You queue an action and a count, and the server works through it online or off: queue → collect the haul → craft it up a chain or sell it → equip or calibrate gear → unlock a harder zone → queue again. A first chain: smith brass, tinker a Brass Spanner, fight Rust Rats. Sixteen skills, each leveled 1 to 100, feed the loop:

- **Gathering:** Prospecting, Lumbering, Scavenging (ore and gems, timber, wreck salvage)
- **Production:** Smithing (plate, blades), Tinkering (tools, rifles, keys), Tailoring (coats, goggles), Cooking (healing rations), Tea Blending (buff teas and tonics)
- **Special:** Calibrating (gear +1 to +20); Alchemy arrives after the MVP
- **Combat:** Vitality, Pressure, Precision, Melee, Fortitude, Marksmanship, Aetherics

Offline play: the Boiler. Logging off leaves 12 hours of steam, shown as a pressure gauge; at zero your Tinker stops until you return. The queue holds two actions ("smelt 400 ingots, then tinker 50 spanners"), and a While You Were Away ledger greets you on return.

Combat. Combat is automatic: pick a zone, a loadout, and triggers ("eat a Ration below 40% HP"), then watch or walk away. Monsters are weak to Melee, Marksmanship, or Aetherics, so loadouts matter. A knockout respawns in 60 seconds and resumes the queue, so offline hours are never wasted.

- **Rustmoor Scrapyards** (1 to 20): Rust Rats; boss Scrapjaw drops Scrapjaw's Cleaver
- **Cinderworks Foundry** (20 to 45): Soot Wraiths; boss Old Bellows drops Bellows Plate
- **Thunderhead Reach** (45 to 70): Storm Harpies; the Lightning Baron, this semester's endgame boss, drops a Coil Staff
- **Dreadnought Hangar** (70+, 5-player dungeon, after the semester): the Iron Dreadnought drops its armor set

Treasure and Calibrating. Monsters also drop Crowns, materials, and ability Schematics. Gear has fixed stats per calibration level, so a +5 Brass Spanner is identical everywhere and fits an order book. Each calibration attempt costs Crowns and materials, with its odds shown up front; failure resets gear to +0 unless a crafted Safety Valve, consumed on use, limits the loss to one level. It is the main Crown sink.

The Brass Exchange (auction house). An order book: sellers list, buyers post standing bids, and matches fill at once, even offline, which suits idle play. Prices move in 2 to 5% steps, a 2% sales tax removes Crowns, and orders expire after 30 days. At launch, NPC sell orders at a price ceiling keep a thin market moving while also removing Crowns.

Guilds and Work Orders. A guild (500,000 Crowns) gets chat, a contribution roster, a Workshop, and 5 weekly Guild Contracts such as "Smelt 50,000 Brass Ingots" whose tiers pay upgrades and tokens. Solo players get a new Work Order every 8 hours; Work Orders and trading unlock at total level 50, so throwaway alt accounts can't farm them.

Gilded Cogs (Stripe). Cogs come in Cog Crates of 10 to open or sell on the Exchange (15% tax), so free players can buy them with Crowns; crates lock 72 hours to blunt card fraud. Cogs buy queue slots, Boiler upgrades (+2 offline hours each, up to 24), cosmetics, and Boiler Boosts; free players can earn all of it through the Exchange.

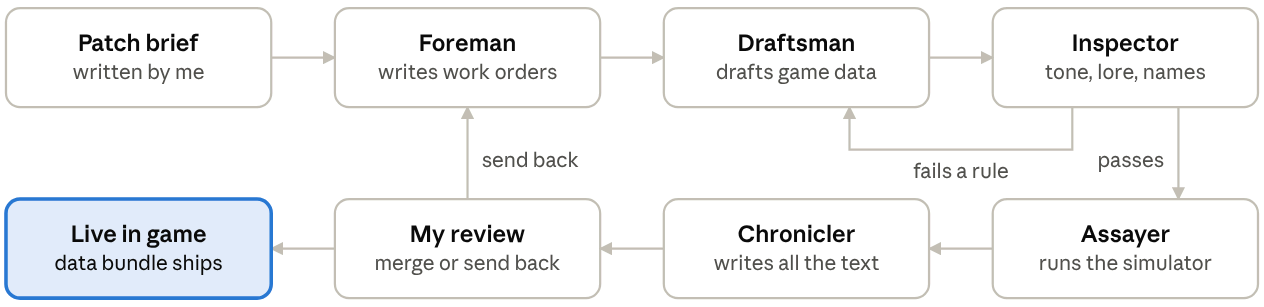
What the player sees. After Google or Discord sign-in, Professor Ottoline Brassbuckle walks you through a copper-to-spanner chain, a first fight, and a first sale. Desktop puts skills left, your Tinker at work center, and chat right; phones use a bottom tab bar. Art is 48×48 pixel sprites in brass, copper, and verdigris; the Tinker wears its gear.

3. AI Architecture

Five Claude agents build and tune content between patches, and each is built around one task it does exceptionally well (one agent, one wow). None sits in a player's moment-to-moment loop: drops, combat, and trades stay in deterministic server code, since idle play runs offline and a traded economy needs repeatable outcomes.

Agent	Its one wow, in one sentence	What players notice
Foreman	Turns a one-page patch brief into a complete, correctly ordered set of work orders.	A new zone or gear tier every two weeks.
Draftsman	Writes new items, recipes, and monsters as game-ready JSON that fits each tier's stat budget.	Thunderhead Reach's monsters and gear.
Inspector	Reviews tone, lore, and names after code has already checked every stat budget.	No off-tone items or overpowered swords.
Assayer	Predicts how a patch will move pacing and prices before players touch it.	About two weeks per zone; stable prices.
Chronicler	Writes the Brass Gazette's prose around facts and opt-in names taken straight from game data.	"Copper up 12%; Iron Owls fell the Baron."

Every patch clears the Inspector and my review before players see it



patch pipeline · 5 agents, 2 checkpoints

One patch (illustrative). For Patch 0.3, Thunderhead Reach, the Inspector bounces the Coil Staff for exceeding its damage budget, and the Assayer finds players reach level 70 in 9 days against a 14-day target, so it proposes cutting Storm Harpy XP by 35%. I approve, merge, and the data ships with no code change.

Pipeline in one sentence. In my game, the Draftsman takes the Foreman's work orders and produces item, recipe, and monster JSON that the player sees as a new zone full of gear to chase.

4. Technical Strategy

One TypeScript codebase on Expo (React Native plus React Native Web) ships the web game now and phone apps later; menus, timers, and sprite sheets need no game engine.

- **Client:** Expo, TypeScript, Reanimated sprites; a PWA on S3 + CloudFront.
- **Server:** authoritative Node.js on ECS Fargate, WebSockets; EventBridge + SQS jobs.
- **Offline:** the server never stops. Each Tinker's state changes through one ordered event log: solo queues resolve on demand, Exchange fills and boss fights write events as they happen, and seeded RNG plus versioned config make every result replayable.
- **Data:** Aurora PostgreSQL for inventory and orders; Valkey for sessions, chat, ranks.
- **Sign-in:** Cognito with Google and Discord (OIDC); 3 Tinkers per account, 1 trading.
- **Payments:** Stripe Checkout webhook to Lambda grants crates once per event ID.
- **Agents:** Step Functions runs each agent as a Lambda calling Claude.
- **Admin console:** an MFA-gated web app to tune drop rates, XP curves, costs, taxes, and the Boiler cap live through a versioned config store, watch economy dashboards, manage players, and approve agent patches, all audit-logged.

Token budget. Each player game action costs 0 tokens, because no agent runs per player. The only daily cost is the Gazette, about 14K tokens or roughly $0.002 a day at Claude Haiku 5.5 pricing, for any number of sessions. Content patches add about 1.15M tokens every two weeks, roughly $3.50 at October 2026 Claude API list prices:

Agent	Model	Calls	Input	Output	Cost
Foreman	Sonnet 5.5	12	120K	40K	$0.64
Draftsman	Sonnet 5.5	9	225K	120K	$1.65
Inspector	Haiku 5.5	12	165K	36K	$0.03
Assayer	Sonnet 5.5	4	120K	40K	$0.64

Agent	Model	Calls	Input	Output	Cost
Chronicler (patch)	Sonnet 5.5	6	48K	40K	$0.50
Chronicler (Gazette)	Haiku 5.5	14	168K	28K	$0.03
Total		**57**	**846K**	**304K**	**$3.49**

Output counts thinking; Draftsman calls include resends. Development adds about 10M Sonnet 5.5 tokens ($36) plus 4 patches ($14), under a $75 hard cap.

Constraints. My biggest constraint is output reliability, because one bad number in a traded economy (an overpowered sword) crashes prices players spent real time on; that is why the Inspector and my review gate every patch.

- **API limits:** per-minute request and token caps mean sequential 10-entity Draftsman batches with backoff; thinking bills as output.
- **Context window:** Sonnet 5.5 holds 1M tokens, but prompts stay near 15K, since Haiku 5.5 costs 5 times more above 100K and smaller prompts stay sharper.
- **Latency:** barely matters; no agent runs while a player waits.
- **Identity and stores:** Discord's Cognito token exchange gets tested in week one, with a Lambda shim as fallback; non-US store billing rules delay app-store builds.

Scope. This semester: 16 skills, 3 zones up to level 70, the Exchange, guilds, leaderboards, Work Orders, the admin console, and a test-mode Cog Shop on web, run by five agents. If time runs short, the Cog Shop's Stripe path drops first, then Guild Contracts, then Discord sign-in. It passes the 48-hour test: two developers could build a rough playable version (one skill chain, the offline timer, one zone, a basic Exchange) in a weekend, and every skill runs on one generic action engine, so the rest is mostly data. Later: the 5-player Dreadnought dungeon, Alchemy, more zones, an Ironclad no-trade mode, live payments, store builds, and a market-watching Warden agent.

Known logic gap. With a small player base the Exchange may stay too thin; NPC sell orders are a start, but the overall Crown faucet and sink budget still needs an owner. The future dungeon's offline rule is already decided: each run snapshots every member's gear and consumables at launch and resolves to the end, so nothing outside the run can change it.

Sources: Claude API pricing · Discord OpenID configuration · Expo: Stripe on iOS